ADHD and Sex Prediction Based on Socio-Demographic and Brain Imaging Data

Farah Alshiha, Alyamama Alsuhaili, Sara Albarghash, Elaf Aljohar, Azzah Alghamdi

Department of Computer Science, College of Computer Science and Information Technology, Imam Abdulrahman Bin Faisal University, P.O. Box 1982, Dammam 31441, Saudi Arabia

{2220007378, 2220001381, 2210009021, 2220004436, 2220006596}@iau.edu.sa

Abstract

Attention Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental condition affecting children and adolescents, yet its diagnosis remains complex due to overlapping behavioral symptoms and sex-based differences. This study aims to develop a predictive model to classify both an individual's sex and ADHD diagnosis using multimodal data, including functional brain imaging, socio-demographic factors, emotional assessments, and parenting information. The proposed methodology integrates machine learning techniques, specifically Artificial Neural Networks (ANNs) for high-dimensional neuroimaging data analysis and Principal Component Analysis (PCA) for dimensionality reduction. A hybrid ensemble approach will also be explored to enhance predictive accuracy and model interpretability. The models will be trained and optimized using Python-based frameworks, mainly PyTorch and Scikit-learn.

Keywords: ADHD prediction, sex prediction, machine learning, fMRI scans, women's health

1. Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder affecting approximately 11% of adolescents, with a higher prevalence in boys (14%)

compared to girls (8%) [1]. However, research suggests that ADHD in females is often underdiagnosed due to differences in symptom presentation [2]. While boys typically display more noticeable hyperactive and impulsive behaviors, girls tend to exhibit inattentive symptoms that may go unnoticed [3]. As a result, undiagnosed ADHD in females can lead to long-term challenges in mental well-being and daily functioning.

This project aims to bridge this diagnostic gap by leveraging machine learning and brain imaging data to enhance ADHD detection. Using functional MRI data, as well as socio-demographic, emotional, and parenting information from the Healthy Brain Network [4], we will develop a predictive model capable of accurately predicting an ADHD diagnosis and the subject's biological sex. Functional MRI data will be analyzed to capture time-series brain activity, generating brain graph matrices that reflect connectivity patterns across different regions.

Our central research question is: "What patterns of brain activity characterize ADHD, and do these patterns differ by sex?" By comparing functional brain connectivity with behavioral observations, we seek to identify sex-specific neurological markers of ADHD. These insights could inform the development of more precise diagnostic tools and tailored treatment approaches, ultimately leading to earlier intervention and improved mental health outcomes. This proposal outlines our plan to develop a multi-outcome model using advanced machine learning and neuroimaging techniques, with the potential to refine diagnostic accuracy and guide future research in ADHD treatment.

2. Review of Related Literatures

Attention Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental disorder affecting children and adolescents, characterized by difficulties with attention, hyperactivity, and impulsivity that can impede day-to-day activities and cause severe impairment and morbidity. Studies suggest that gender differences play a key role in how the condition presents, is diagnosed, and is treated. Males are diagnosed more often, mainly

because they exhibit more externalizing behaviors, while females tend to have internalizing symptoms, which can lead to underdiagnosis.

De Rossi et al. conducted a retrospective observational study examining gender-related clinical characteristics in ADHD [1]. The study included 715 children and adolescents (mean age = 9.4 years) diagnosed between 2018 and 2020 at Bambino Gesù Children's Hospital in Italy. Results showed that boys had higher IQ scores than girls, but both groups experienced similar levels of functional impairment and adaptive skills. Girls displayed more severe ADHD symptoms, suggesting that those who are clinically diagnosed may present with more pronounced symptoms, likely due to delayed recognition of their condition. In contrast, boys showed greater tendencies toward withdrawn and depressed behaviors, internalizing symptoms, and obsessive-compulsive traits, indicating a possible link between ADHD in boys and mood disorders [1].

The study also highlighted a referral bias, where girls were often diagnosed only when their symptoms were more severe, particularly because of their predominantly inattentive presentation. This pattern may contribute to underdiagnosis and delayed treatment interventions. While boys are generally associated with externalizing behaviors, the study found that they also exhibit significant internalizing symptoms, which are frequently overlooked. These findings emphasize the need for a more thorough assessment process for boys and earlier interventions for girls. Treatment should prioritize managing inattention in girls and addressing internalizing symptoms in boys with cognitive-behavioral strategies. The study concluded that, in clinical settings, girls with ADHD tend to present with more severe symptoms and lower IQ scores, while boys show higher levels of internalizing behaviors and obsessive-compulsive tendencies [1]. Recognizing these gender differences is crucial for improving diagnosis and treatment approaches.

Adult ADHD prevalence is higher in males than females, and younger adults show greater prevalence than older age groups. Research indicates that adults with ADHD are less likely to be employed, have lower educational levels, and tend to lead unhealthy lifestyles,

including poor eating habits and sleep disturbances. Additionally, parental trauma can negatively impact children's mental health, affecting psychiatric outcomes and increasing the risk of ADHD [2]. This study focused on university graduates to explore the link between ADHD traits and sociodemographic factors among 1,240 workers in Japan. Participants completed an internet-based questionnaire gathering information on their sociodemographic characteristics, such as gender, income, age, health-related behaviors, working time, physical activity, eating habits, and sleeping hours. Results indicated that men had higher ADHD traits compared to women, and younger individuals had higher trait levels. Workers with lower incomes displayed more ADHD traits, and a greater frequency of midnight meals and insufficient sleep were linked to higher ADHD traits. The study also found a connection between ADHD and smoking, as well as an association between ADHD symptoms and frequency of drinking among college students [2]. Recognizing these connections is necessary for developing targeted interventions that consider the differences among individuals with ADHD traits.

Furthermore, it has been discovered that ADHD is not related to age nor developmental process directly [3]. This study tries understanding the risks of ADHD in children and how it is related to parents' demographics, particularly education level, marital status, health, and age. A survey was conducted to classify and assess respondents into specific criteria followed by an analysis of children's ADHD. The study was structured into three parts: 1) characterizations of respondents 2) children's profile 3) sociodemographic profile of mothers of children with ADHD risks [3]. Key findings of this study present parental education play a role, mothers with low education levels are more likely to have children with high ADHD risk whereas high education mothers showed more awareness and provide early intervention to their kids.

ADHD does not only affect children but also influence the way parents manage their children, since it is challenging to raise a child with impulsivity, hyperactivity, and inattention. Such parenting styles can lead to secondary development of co-morbid psychiatric and behavioral problem, making managing ADHD more complex [4]. Parents

of ADHD children, specifically mothers, tend to quit or work less hours believing their children need more assistance and support. However, due to these symptoms these parents may become emotionally less engaging, responding with higher levels of verbal aggression and stricter disciplinary actions for troubled behaviors. This reaction can make ADHD children behave more negatively [4].

To address these challenges behavioral parent training protocols have been developed for parents of children with ADHD to help adopt more effective parenting styles and focus on the positive techniques. The study [4] is a cross sectional on 48 children diagnosed with ADHD. Semi structured interviews were done for the child and one or both parents to gather detailed behaviors and family background. Assessments tools that were used are: the Arabic version conner's parent rating scale for parents to detect whether their child has ADHD, the Stanford Binet intelligence scale 5th edition that was designed to test the cognitive abilities and its ability to provide a full scale IQ, and a parenting style questionnaire perceived by children. The collected data were digitized and analyzed to determine the correlation between sociodemographic factors and ADHD severity [4].

With an emphasis on sex and gender disparities in screening and diagnostic history, Assari's 2023 study [5] carried out a comprehensive investigation on the emotional, behavioral, and cognitive aspects of ADHD. Using information from the Adolescent Brain Cognitive Development (ABCD) study, which involved 10,171 American children between the ages of 9 and 10, the study looked at the differences between kids who had previously been diagnosed with ADHD and kids who tested positive in a community context. The purpose of the study was to ascertain if the observed variations in ADHD traits between males and females are due to biological sex differences or are impacted by biases in diagnosis and referral based on gender. The results showed that lower cognitive performance and higher levels of internalizing, externalizing, and overall behavioral issues were linked to both positive screening and diagnosis histories. Significant sex differences, however, were mostly found in the diagnostic group, where females showed more severe cognitive deficits and males showed larger connections with externalizing behaviors.

Even after controlling for sociodemographic factors such family structure, parental education, and household income, differences in referral and diagnostic procedures remained. The findings imply that rather than being biologically driven, some of the previously believed sex-based disparities in ADHD may be socially produced. The study's primary drawback was its dependence on parental reports for behavioral assessments and ADHD diagnosis, which could introduce bias. To avoid potential biases in treatment and intervention tactics, Assari (2021) emphasizes the significance of differentiating between sex and gender variations in ADHD research and calls for more fair screening and diagnostic practices. Ultimately, the study provides crucial insights into how diagnostic pathways may contribute to disparities in ADHD presentation and management, emphasizing the need for gender-sensitive approaches in clinical and educational settings.

By examining 5,777 multimodal predictors spanning demographics, neuroimaging, and psychosocial factors, Lacy et al.'s study used machine learning to forecast the onset of ADHD using data from the ABCD study [6]. With 86.1%–93.7% accuracy and AUROC values ranging from 0.919 to 0.985, multimodal models outperformed neural-only models in terms of predictive performance. The results highlighted the importance of psychosocial factors in predicting the start of ADHD, with excessive sleepiness, parent-child conflict, and previous mental health treatment standing out as important predictors. There may be a protective effect because academic success, especially getting good grades, was inversely correlated with ADHD. The prediction value of neural-only models was lower, highlighting the significance of including behavioral and environmental evaluations. Furthermore, whereas structural MRI measures played a role in the models, psychosocial variables had a greater impact. The study emphasized the significance of multifaceted approaches in early ADHD diagnosis and the need for real-world validation and practical implementation of predictive frameworks. To enhance clinical utility in a variety of groups and improve predictive models, future studies should investigate external validation in greater detail.

In Garcia-Argibay et al.'s [7] 2022 study, the use of a variety of machine learning models, such as logistic regression, random forest, XGBoost, and deep neural networks (DNN), the study examined Swedish registry data from 238,696 people to forecast the onset of ADHD. With an AUROC of 0.75 and a balanced accuracy of 0.69, the DNN model performed better than the others. Male sex, scholastic failures, a family history of ADHD, having a parent with criminal convictions, and speech or learning difficulties were all important predictors of ADHD. These results demonstrate the significance of both environmental and genetic factors in the development of ADHD. The study emphasized the need to integrate behavioral and psychosocial elements for better predictive accuracy, even though brain models contributed. According to the research, machine learning models may help physicians identify ADHD early, reduce misdiagnosis, and enhance treatments techniques if they are externally verified. To improve clinical applicability, future research should concentrate on improving models by adding more predictors and confirming results across a range of groups.

Traditionally, it is diagnosed based on Psychiatric interviews, and behavioral assessments but these usually tend to be time-intensive and often subjective [5]. Moreover, long waiting times for assessment (up to 3 years) put a strain on the National Health Service (NHS) and increase economic burdens (€17,769 per person annually in the UK) [10]. With the increasing demand for early diagnosis, researchers explored approaches offered by Artificial Intelligence trying to improve efficiency and accuracy, focusing on neuroimaging (brain imaging), and physiological signals [4][5][10].

Brain Imaging Imposes Insights into Differences in the Structure and Function of ADHD Patients [5] individuals with ADHD showed lower cortical surface area and thickness, a disruption in FC, as well as decreased brain volume [4]. Functional MRI (fMRI) studies have shown abnormalities in the Default Mode Network (DMN), that may underlie inattention and impulsivity [4]. Within the Neuroimaging-based diagnosis a hybrid AI system, NeuroIntel, was developed and trialed to excel in the NHS using a combination of

machine learning (ML) models and knowledge-based systems to improve ADHD diagnosis [10].

Electroencephalogram (EEG) is commonly used in the ADHD area of research as a non-invasive Physiological Signal-Based method for recording electrical activity of the brain. In EEG Data, AI models provide >80% accuracy, differentiate ADHD patients. In Heart Rate Variability (HRV) which is another Physiological Signal-Based approach ML models were used on electrocardiogram (ECG) data, with a classification accuracy of 87.2% ADHD [5]. AI-Based Questionnaire Analysis is an Emerging Diagnostic tool where diagnosis rely on Conners' Rating Scales, Adult ADHD Self-Report Scale (ASRS), and the Diagnostic Interview for ADHD (DIVA). AI techniques are used to automate questionnaire analysis, with decision tree and random forest classifiers achieving accuracies up to 100% [5][10].

A decision tree algorithm was selected due to its high interpretability, thus making it best suited for clinicians. The study confirmed that the model was 75.03% accurate when all features, including DIVA attributes, were used. Dropping DIVA attributes resulted in a lower accuracy of 65.27%. Notably, the hybrid approach, which integrated machine learning to a knowledge-based strategy, significantly enhanced decision support to a remarkable 93.61% accuracy. This success proves the power of merging vary ing methodologies with clinical decision-making [10]. Game Simulations where Virtual reality-based tests (e.g., Continuous Performance Test [CPT], Reverse Stroop Task [RST]) provide an interactive assessment of attention and impulsivity. AI applied to game data achieved 78% accuracy, potentially providing a child-friendly alternative to diagnosis [5].

Wearable Devices and Motion Data as a Diagnostic Tool, an accelerometer-based CNN model was generated that achieved an accuracy of 98.6%. Wearable technologies (e.g., smartwatches) may facilitate real-time ADHD tracking and remote assessment [5].

Several significant challenges and future directions ADHD research field is facing today: there are no publicly available datasets, and most of them come from private sources, which do not allow reproducing studies. Furthermore, there is an urgent need for explainable AI models, since several deep learning models are known to act as "black boxes," limiting clinician trust. In addition, future investigation should also focus on a multimodal data-type integration of neuroimaging, physiological signals, behavioral tests, and genetic information to increase accuracy. Finally, exploring wearable and telehealth solutions, such as ECG, PPG, and motion sensors, could revolutionize real-time monitoring and management of ADHD [5].

However, the utilization of AI in ADHD diagnostics provides exciting opportunities for improvement in both accuracy and efficiency. While Neuroimaging remains a powerful tool, almost all physiological signals, behavioral data, and wearable technologies can provide low-cost alternatives. As this technology continues to develop, AI-driven products may become an integrated part of the ADHD assessment and management process, helping to considerably minimize diagnostic timeframes and elevate overall patient wellbeing.

The current body of knowledge considers ADHD in both males and females to be of the same nature in their focus of study and seldom take the time to point out the drastic differences that would deem them two separate subfields of research. Most studies present diagnostic criteria based on the male representation of ADHD. These studies, in turn, greatly hinder diagnosis predictions made with females, leading to an inaccurate representation of what ADHD in females may look like [8]. Moreover, this field of research lacks longitudinal studies that track ADHD symptoms in females over time, failing to add a wider lens to the general, bigger-picture idea of how this disorder takes shape during a lifespan, and not merely a small window of time [9].

In another sense, when research is done on females with ADHD, most studies often focus on study subjects of predominantly European ancestry and fail to include racial minorities.

This may be due to the setting in which most of these studies are conducted, in which any race that is not white is a minority in the literal sense. This is greatly concerning when it comes to diagnosing ADHD in females of color, as it is the least likely focus of study of all general demographics [10].

This paper aims to develop the body of knowledge regarding ADHD symptoms in females but with the use of AI techniques, namely methodical feature engineering within the neural network design. Recently, AI has been used to take in existing scientific knowledge and add creativity to discover new facets about these already existing pieces of information. The implementation of this research endeavor is all to better understand and further Women's brain health. The general gaps mentioned above can be addressed building on the findings of this paper, yielding another lens to this already obscure field of study.

Table 1: Summary of Relevant Findings

MINISTRY OF EDUCATION
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
COLLEGE OF COMPUTER SCIENCE
جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Ref.	Title	Year	Dataset	ML	Result	Notes
13	ADHD Among a Cohort of Ethnic Minority Women	2010	-	-	Increased impairment was linked to women with attention deficit-hyperactivity disorder who also had comorbid psychological problems, such as significant sadness and anxiety.	To support successful therapies and recovery, broader standards that show fair evaluation and identification of attention deficit-hyperactivity disorder symptoms for both sexes are necessary.
12	Females with ADHD: An expert consensus statement taking a lifespan approach providing guidance for the identification and treatment of attention-	2020	-	-	The potential challenges of working with this patient group were identified, as well as specific barriers that may hinder recognition.	-

	deficit/ hyperactivity disorder in girls and women					
5	Emotional, Behavioral, and Cognitive Correlates of Attention Deficit and Hyperactive Disorder (ADHD) Screening and Diagnosis History: Sex/Gender Differences	2021	From the ABCD Study: sample of 10,171 American children between ages 9 and 10 years old. This sample included 1,488 children with a history of psychiatric diagnosis and 8,683 children without a diagnosis.	-	The study found that poorer cognitive performance, higher internalizing, externalizing, and total problem behaviors, higher inattention, and lower BMI were linked to positive ADHD screening and a history of diagnosis, with sex not affecting these relationships.	No participants were excluded for any reason other than missing data
8	A Biased Perspective on	2021	No new dataset; the paper reviews existing	-	Critical analysis of biases in ADHD brain	Emphasizes the need for improved data-

	Brain Imaging of ADHD		brain imaging studies on ADHD.		imaging studies, highlighting inconsistencies and limitations.	sharing, better diagnostic consistency, and cautious interpretation of neuroimaging findings.
4	Socio-demographic and clinical correlates of parenting style among parents having ADHD children: a cross-section study	2021	The study involved 48 children diagnosed with ADHD, aged 6 to 12 years, along with their parents. Assessments included the Stanford-Binet Intelligence Scale (5th edition), Conner's Parent Rating Scale-Revised, a parenting style questionnaire, and the Fahmy and El-Sherbini	-	Findings indicated that mothers exhibited significantly lower over-protection scores compared to fathers. Positive parenting styles were more prevalent among parents of children with mild ADHD symptoms, higher social functioning, and those from higher socioeconomic	The study underscores the influence of sociodemographic factors on parenting approaches in families with ADHD children, highlighting the need for tailored interventions that consider these variables.

			socioeconomic status measure.		backgrounds. Additionally, a positive correlation was found between children's IQ and maternal warmth/support.	
1	Gender-Related Clinical Characteristics in Children and Adolescents with ADHD	2022	The study retrospectively analyzed data from 715 children and adolescents (ages 6–18) diagnosed with ADHD between 2018 and 2020 at a single center. Assessments included the Child Behavior Checklist (CBCL) and the Conners' Parent Rating Scale-Revised: Long (CPRS-R:L)	-	The study found that girls had more severe ADHD symptoms and lower IQ scores, while boys had higher levels of internalizing problems and obsessive-compulsive symptoms.	The study underscores the importance of considering gender-specific characteristics in ADHD treatment, as girls may exhibit more severe symptoms upon referral.

9	Automated detection of ADHD: Current trends and future perspective	2022	-	The paper explores the use of machine learning and deep learning techniques in ADHD diagnosis, highlighting gaps such as limited dataset availability and limited use of wearable device data.	The review emphasizes the potential of artificial intelligence in improving ADHD diagnosis, suggesting the need for more publicly available datasets and increased research on wearable device data.	The authors advocate for AI-driven clinical decision support systems that integrate multiple diagnostic tools to enhance early and accurate ADHD diagnosis in clinical and non-clinical settings.
7	Predicting childhood and adolescent attention-deficit/hyperactivity disorder	2022	Registry data from 238,696 persons born and living in Sweden between 1995 and 1999	logistic regression, random Forest, gradient boosting,	The DNN model was the most fitting, with a balanced accuracy of 0.69, sensitivity of	This research underscores the promise of integrating advanced computational

	onset: a nationwide deep learning approach			XGBoost, penalized logistic regression, (DNN), and ensemble models	71.66%, and specificity of 65.0% at the 0.45 probability threshold.	techniques with clinical data to improve early detection and intervention strategies for ADHD
2	Associations Between Adult Attention-Deficit/Hyperactivity Disorder (ADHD) Traits and Sociodemographic Characteristics in Japanese Workers	2023	1240 workers from different regions of Japan who had graduated from university. The study utilized an adult ADHD Self-report Scale to measure ADHD traits, focusing on sociodemographic factors such as sex, age, socioeconomic status, working time, and health-related behaviors.	-	The study found that males, younger individuals, low-income workers, frequent midnight meal consumption, and lack of sleep were associated with higher ADHD traits, with no differences observed for breakfast, lunch, and dinner.	The study advances our knowledge of the relationship between sociodemographic characteristics and ADHD features in the Japanese workforce.

11	Diagnosing attention-deficit hyperactivity disorder (ADHD) using artificial intelligence: a clinical study in the UK	2023	Existing literature on adult ADHD diagnosis in women	-	A large number of women have undiagnosed ADHD, which poses serious obstacles in both their personal and professional lives. Getting a diagnosis as an adult frequently led to better self-awareness and easier access to the right treatments.	The study emphasizes the necessity for increased awareness and tailored diagnostic criteria to better identify and support women with ADHD, addressing the historical underrepresentation and misdiagnosis in this population
10	Diagnosing attention-deficit hyperactivity disorder (ADHD) using artificial intelligence: a clinical study in the UK	2023	501 anonymised patient data from the Adult ADHD Service of the National Health Service (NHS) in the United Kingdom, gathered until July 2022. Validated	Researchers developed a hybrid AI algorithm that combines machine learning models	The hybrid model achieved an accuracy of 93.61% in diagnosing ADHD when all features, including the Diagnostic Interview for ADHD in Adults	The study suggests AI-assisted ADHD diagnosis can be expedited, especially in limited specialist access, and can improve diagnostic precision

			clinical diagnostic interviews and screening questionnaires were among the data.	with knowledge-based systems to aid in diagnosing adult ADHD, using decision tree algorithms for predictive modeling.	(DIVA), were used. Without DIVA attributes, the machine learning model alone reached an accuracy of 65.27%.	without the DIVA assessment, reducing medical resource strain.
6	Identifying selective predictors of ADHD, Oppositional Defiant and Conduct Disorder onset in early adolescence with optimized deep learning	2023	5,777 multimodal predictors of children ages 9-10 from the ABCD cohort	Deep learning optimized with an innovative AI algorithm for early prediction of ADHD, ODD, and CD	High predictive accuracy (86-97%), AUROC (0.919-0.996), precision, and recall (82-97%)	Psychosocial factors were more helpful than cognitive or physiological measures, while structural MRI metrics in the limbic system showed potential for predicting the development of CD.

3	The Population of Children with High Risk of ADHD and The Mother's Sociodemographic Profile in Surabaya	2024	Data from 358 mothers of children in grades 3 to 6 at three elementary schools in Surabaya.	-	15.1% of children showed signs of ADHD risk based on the SPPAHI questionnaire.	The study emphasized the need for further research to diagnose ADHD accurately in children.

3. Project Deliverables of the team

This section shows the deliverables of the project.

Table 2: Deliverables

Deliverable	**To whom**	**Delivery Media**	**Duration**	**Date**
Literature Review (Homework-1)	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	1 week	February 23rd, 2025
Project Proposal	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	5 days	March 2nd, 2025
Project Proposal Presentation	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	5 days	March 2nd, 2025
Description of Selected ML Algorithms	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	1 week	March 25th, 2025
Final Project Report	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	2 weeks	May 4th, 2025
Final Project Presentation	Dr. Rabab Ahmad Abduallah Alkhalifa	Softcopy	2 weeks	May 11th, 2025

4. Description of the Proposed Techniques

4.1 Technique #1: Random Forest with MultiOutput Wrapper

A Random Forest builds many decision trees on bootstrap samples and random feature subsets, then averages their votes for classification. Since our target has multiple binary labels, we use MultiOutputClassifier(RandomForestClassifier(...)) so that one forest is trained per label in parallel. This approach is well-suited to our biomedical data because it handles both numerical and categorical variables with minimal preprocessing, is robust to noise and outliers, and produces feature importance scores that show which connectivity measures and metadata features most determine each ADHD subscale. For more details refer to https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html [14]

4.2 Technique #2: XGBoost (Extreme Gradient Boosting)

An ensemble of decision trees is constructed stage-by-stage by the efficient gradient-boosting framework XGBoost. The objective function uses L1/L2 regularization to control complexity, and new trees are fitted to the current ensemble's residual errors at each step. Internally, it employs parallelized tree construction to increase training speed and a sparsity-aware algorithm for effective split detection on high-dimensional data. For more details refer to https://xgboost.readthedocs.io/en/release_3.0.0/ [15]

4.3 Technique #3: ANN (Artificial Neural Network)

An input layer, one or more hidden layers with non-linear activation functions, and an output layer make up an ANN, a feed-forward neural network. The model can learn intricate, non-linear interactions by optimizing network weights using backpropagation to minimize a selected loss function. The design can be modified to fit the task's complexity, which is determined by the number of layers and neurons. For more details refer to https://pytorch.org/tutorials/beginner/basics/buildmodel_tutorial.html [16]

This study seeks to address aforementioned gaps within the literature by developing a machine learning model capable of predicting both an individual's sex and their ADHD diagnosis using a combination of functional brain imaging data and socio-demographic, emotional, and parenting information. This is feasible through an artificial neural network that uses gradient descent in backpropagation as its main algorithm. We will employ a mutli-label binary classification approach, wherein the model will output two labels, both of which are binary (Female/Male and ADHD/no AHDH).

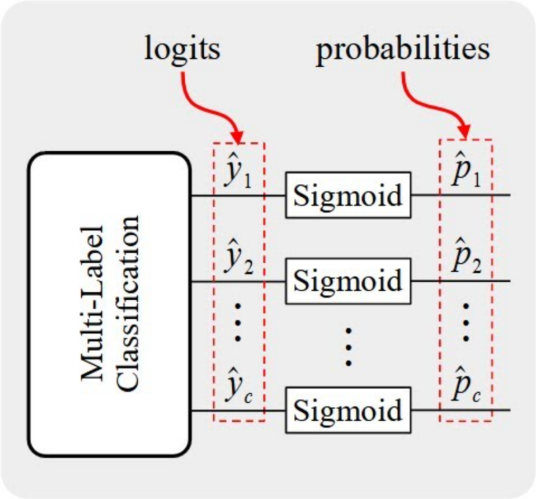


Figure 1: Multi-label binary classification

The figure above demonstrates the general architecture of the model. The input will consist of MRI brain imaging data alongside categorical information that is appropriately encoded. The brain imaging data spans thousands of features. That is why it is crucial that this data specifically undergoes a form of dimensionality reduction, like PCA.

Following the fully connected layer that contain within them the neural network embeddings are two simple logistic regression models that employ the Sigmoid function. This is similar to using logistic regression as the output layer, but we designed the model in such a way as to train the two variables separately for more control. Depicted below is the sigmoid function plotted on a graph.

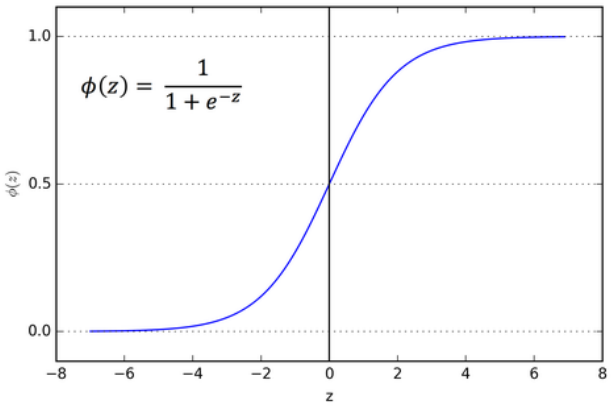


Figure 2: Sigmoid Function in Logistic Regression

We are using the Sigmoid function, as opposed to other functions like Tanh, because both of our predictions range from 0 to 1. The reason for the use of logistic regression lies in its ability to offer better interpretability for clinical studies. It is also useful in providing more insight into ADHD diagnoses through its reported feature coefficients. The output of the artificial neural network model will contain two heads, one for sex prediction and the other for ADHD prediction. After the final fully connected layer, the representation will be propagated forward to two logistic regression heads. This is discussed in further detail in the next section. The reason for the integration of both the ANN and Logistic Regression model is in the hope that it will yield a powerful hybrid model that will output with better performance and interpretability than if they were used separately.

5. Empirical Studies

5.1 Description of dataset

Table 3: Datasets of the Study

Table	Shape	description
Functional Connectome Matrices	(1213, 19901)	MRI connectome matrices.
Categorical Metadata	(1213, 10)	Bratt Simplified Measure of Social Status, Pre-Interview-Demographics/Family, and MRI Information
Quantitative Metadata	(1213, 19)	Alabama Parenting Questionnaire, Strength and Difficulties Questionnaire, and Laterality Index Score
Training Solution	(1213, 3)	ADHD Outcome and Sex

5.1.1 Statistical Analysis of the Dataset

The statistical analysis of the dataset is presented in Table 4 below. The mean, median, standard deviation, quarter quantiles, maximum and minimum values of the dataset are presented.

Table 4: Statistical Analysis of the Quantitative Dataset

	count	mean	std	min	25%	50%	75%	max
EHQ_EHQ_Total	1200.0	59.514750	49.744631	-100.0	46.670000	77.800000	94.470000	100.000000
ColorVision_CV_Score	1190.0	13.419328	2.112830	0.0	14.000000	14.000000	14.000000	14.000000
APQ_P_APQ_P_CP	1201.0	3.819317	1.330348	2.0	3.000000	3.000000	4.000000	12.000000
APQ_P_APQ_P_ID	1201.0	13.337219	3.593562	5.0	11.000000	13.000000	16.000000	28.000000
APQ_P_APQ_P_INV	1201.0	39.767694	4.872729	23.0	36.000000	40.000000	43.000000	50.000000
APQ_P_APQ_P_OPD	1201.0	17.963364	3.333965	7.0	16.000000	18.000000	20.000000	28.000000
APQ_P_APQ_P_PM	1201.0	16.557036	5.146548	5.0	13.000000	16.000000	19.000000	37.000000
APQ_P_APQ_P_PP	1201.0	25.498751	3.053720	13.0	23.000000	26.000000	28.000000	30.000000
SDQ_SDQ_Conduct_Problems	1204.0	2.074751	2.022992	0.0	0.000000	2.000000	3.000000	10.000000
SDQ_SDQ_Difficulties_Total	1204.0	12.213455	6.517330	0.0	7.000000	12.000000	17.000000	34.000000
SDQ_SDQ_Emotional_Problems	1204.0	2.324751	2.166399	0.0	1.000000	2.000000	4.000000	10.000000
SDQ_SDQ_Externalizing	1204.0	7.613787	4.130399	0.0	5.000000	8.000000	10.000000	20.000000
SDQ_SDQ_Generating_Impact	1204.0	4.103821	2.808079	0.0	2.000000	4.000000	6.000000	10.000000
SDQ_SDQ_Hyperactivity	1204.0	5.539037	2.807063	0.0	4.000000	6.000000	8.000000	10.000000
SDQ_SDQ_Internalizing	1204.0	4.599668	3.510469	0.0	2.000000	4.000000	7.000000	17.000000
SDQ_SDQ_Peer_Problems	1204.0	2.274917	2.089041	0.0	0.000000	2.000000	4.000000	9.000000
SDQ_SDQ_Prosocial	1204.0	7.740864	2.094832	0.0	6.000000	8.000000	10.000000	10.000000
MRI_Track_Age_at_Scan	853.0	11.245678	3.234372	0.0	8.803901	10.739219	13.460871	21.564453

5.1.2 Correlation of the Dataset

The correlation between each feature pair of the categorical dataset is presented in Figure 3 below. The heatmap displays which pairs of features have the strongest correlation in association with the color scale.

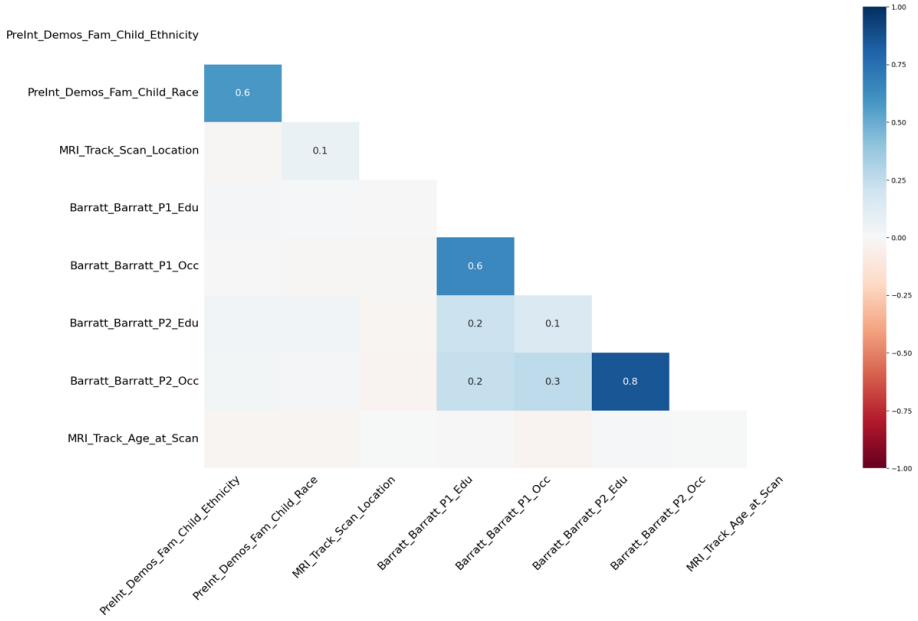


Figure 3: Heatmap of Categorical Dataset

5.1.3 Functional Connectivity

An example of an fMRI sample from the functional connectome matrices dataset can be found in Figure 4 below. Connected nodes, which represent strong correlations between regions of the brain, signify a threshold of 0.5 or greater.

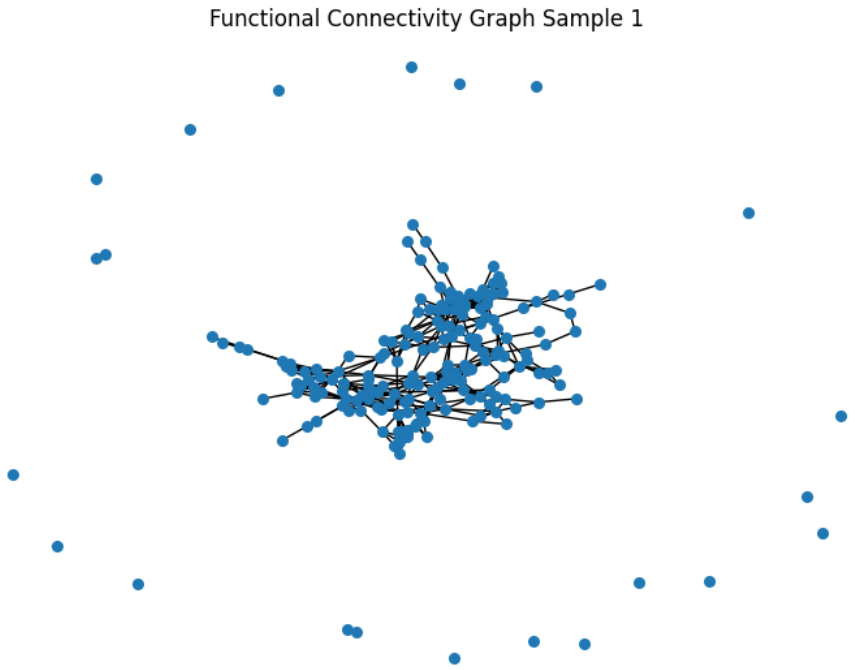


Figure 4: Functional Connectome matrix for a single anonymous participant

5.2 Experimental Setup

All experiments were conducted in a Python 3.10 Jupyter Notebook. Key libraries included pandas and NumPy for data handling, scikit-learn (v1.x) for preprocessing, partitioning, and classical classifiers, XGBoost for gradient boosting, and PyTorch for the multi-task neural network.

All tables were read via pd.read_csv or pd.read_excel. To start off, categorical and quantitative metadata were merged on participant_id (inner join). The resulting table was then joined with the functional connectome on participant_id, and the identifier column was dropped—yielding a feature matrix of size N×D (D≈19927).

To match prior WiDS baselines, we applied a single hold-out split:

X_train, X_val, y_train, y_val = train_test_split(x, y, test_size=0.20, shuffle=True, random_state=42)

This produced an 80% training set, and 20% validation set. All random seeds (random_state) were fixed to 42 to ensure reproducibility.

Preprocessing and dimensionality reduction:

- Scaling: Features in X_train were standardized to zero mean and unit variance using StandardScaler().fit_transform; the same transformation was applied to X_val via sc.transform.
- PCA: To reduce dimensionality and noise, we fit a PCA model on the scaled training data with n_components=0.95 (retaining 95% variance), then projected X_val accordingly. This reduced D from 19927 to ≈750 principal components.

For each of four methods, we trained two variants—one on the full scaled data and one on the PCA-reduced data—using a MultiOutputClassifier to handle the two binary targets (ADHD outcome and sex). Hyperparameters were chosen to mirror practices in the neuroimaging classification literature and the WIDS baseline code.

Table 5: Model Parameters

Model	**Key Parameters**
Random Forest	RandomForestClassifier(criterion='log_loss', random_state=42)
Gradient Boost	GradientBoostingClassifier(loss='log_loss', learning_rate=0.01, random_state=42)
XGBoost	XGBClassifier(objective='binary:logistic', eval_metric='logloss', random_state=42)
Neural Network	Custom MultiTaskANN—one shared FC layer (128 units, ReLU), dropout=0.3; two sigmoid heads; optimizer=Adam(lr=0.001); batch_size=32; epochs=20

Each classical model was trained via .fit(X_train, y_train); the neural network was trained in mini-batches with binary cross-entropy loss summed over both heads.

After training, predictions on the hold-out set were obtained via .predict(X_val) (or via the NN's forward pass), and we reported precision, recall, and F1 -score using classification_report. To facilitate replication, we exported the final merged training and validation tables to CSV (merged_train_ML.csv, merged_val_ML.csv) and set all random seeds explicitly to 42.

A schematic diagram of this pipeline—from data ingestion through model evaluation—is provided in Figure 2 (flowchart), and the full code is available in the accompanying notebook.

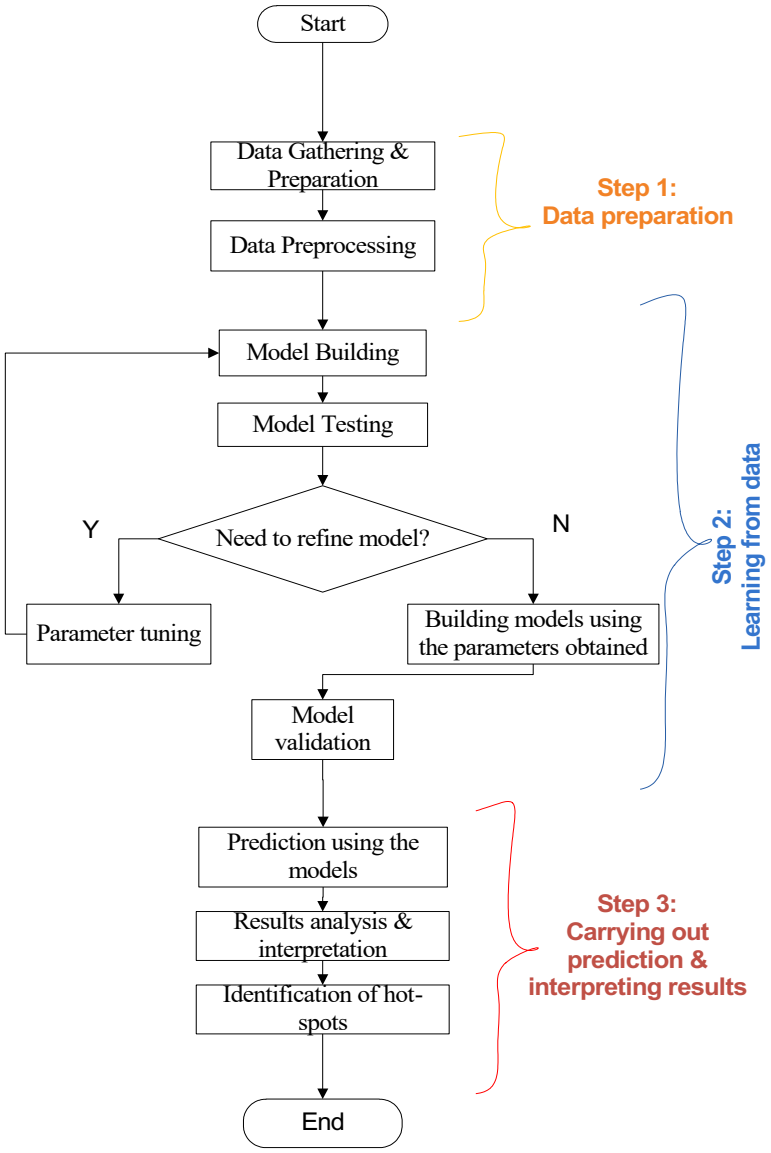


Figure 5: Machine Learning Process Flowchart

5.3 Performance Measures

The classification_report function in the sklearn.metrics module produces a compact summary of a classifier's performance by comparing true labels (y_true) to predicted labels (y_pred). You can optionally specify which classes to include (labels/target_names),

control numeric formatting (digits), handle zero-division cases (zero_division), weight samples (sample_weight), or get the results back as a dictionary (output_dict=True).

By default, it reports, for each class, precision, recall, F1-score and support (number of true instances), plus overall averages—macro (unweighted mean), weighted (support-weighted mean), and micro (aggregate true/false counts, shown primarily for multilabel tasks). In binary settings, recall for the positive class is sensitivity and for the negative class is specificity.

5.4 Optimization strategy

The ANN employed a multi-task learning framework with two binary output heads—one for ADHD diagnosis and the other for sex classification—sharing a common hidden representation. Optimization of the model parameters was carried out using the Adam optimizer, a widely used gradient-based method that combines the benefits of momentum and adaptive learning rates. Adam was selected for its robustness and efficiency when training deep learning models on high-dimensional and potentially sparse datasets such as ours.

The network minimized a binary cross-entropy loss, suitable for each binary classification task. Initially, the loss was computed by combining the two outputs into a joint prediction; however, this approach risked uneven learning if one task dominated the gradients. To address this, the final implementation calculated the loss for each task independently and averaged them, ensuring a more balanced optimization process across both outputs.

Regularization was introduced through a dropout layer (p=0.3) after the shared hidden layer, helping mitigate overfitting by randomly deactivating a fraction of neurons during training. The model was trained using mini-batch gradient descent with a batch size of 32 and a fixed learning rate of 0.001 over 20 epochs. These hyperparameters were selected based on empirical performance and prior literature but could be further refined via systematic hyperparameter tuning. No explicit learning rate scheduler or early stopping

mechanism was used, though these could be explored in future iterations to dynamically adjust the learning rate based on validation performance.

6. Result and Discussion

Presented below are the results and discussions of various experimental options. With PCA applied to the full feature set, XGBoost slightly outperforms Random Forest achieving a precision of 0.66 versus 0.65, perfect recall of 1.00 versus 0.94, and a higher F1 score (0.80 vs. 0.77).

Table 6: Results of using the complete features with PCA

Quality measures	RandomForest	XGBoost	ANN
Precision	0.65	0.66	0.72
Recall	0.94	1.00	0.46
F1 Score	0.77	0.80	0.56

Table 7: ANN Classification Report

ADHD Classification report				
	Precision	**Recall**	**F1-score**	**Support**
No ADHD	0.24	0.31	0.27	65
ADHD	0.72	0.65	0.68	178
Accuracy			0.56	243
Macro avg	0.48	0.48	0.48	243
Weighted avg	0.59	0.56	0.57	243
Sex Classification Report				
	Precision	**Recall**	**F1-score**	**Support**
Male	0.70	1.00	0.83	171

Female	0.00	0.00	0.00	72
Accuracy			0.70	243
Macro avg	0.35	0.5	0.41	243
Weighted avg	0.50	0.70	0.58	243

The multitask artificial neural network demonstrated uneven performance across the two classification tasks of ADHD diagnosis and biological sex prediction, as shown for reference in Table 5. For the ADHD classification task, the model achieved a weighted F1-score of 0.57 and an overall accuracy of 56%. Notably, the model showed considerably higher precision and recall for the ADHD class (precision = 0.72, recall = 0.65) compared to the No ADHD class (precision = 0.24, recall = 0.31). This suggests a bias toward positive ADHD classification, which may be reflective of either a class imbalance in the training data or stronger signal features for ADHD presence.

In contrast, performance on the sex classification task was markedly poorer. The model achieved perfect recall for the male class (1.00), but completely failed to identify any female samples, with zero precision, recall, and F1-score for that class. This outcome implies that the model only predicted the male class, resulting in an inflated overall accuracy of 70% that is misleading given the complete lack of sensitivity to the female class. The macro-averaged F1-score of 0.41 more accurately reflects the model's poor balance and generalizability in this task.

These findings indicate that while the model captures some discriminative patterns related to ADHD, it struggles significantly with sex classification. The results also underscore the potential impact of class imbalance, particularly for the sex classification task where the model likely encountered fewer female samples during training. Furthermore, the network may be overfitting to dominant class features, warranting the application of loss weighting, data augmentation (e.g., SMOTE), or resampling techniques to improve minority class detection. Additionally, further model calibration, threshold tuning, or architectural adjustments may be necessary to improve multi-task performance parity.

6.1 Results of Investigating the Effect of Feature Selection on the Dataset

In an effort to manage the high dimensionality of the dataset—comprising approximately 19,920 features primarily derived from flattened functional connectome matrices—we implemented Principal Component Analysis (PCA) as a dimensionality reduction strategy prior to model training. The goal was to reduce computational complexity, limit overfitting, and improve model generalization.

However, our results indicate that PCA negatively impacted the model's performance, particularly in the context of the multi-task classification problem involving ADHD diagnosis and sex prediction. While PCA efficiently reduced the number of input features by projecting them onto a set of orthogonal principal components that retained 95–99% of the original variance, it failed to preserve the non-linear and fine-grained structures inherent in the original data. Functional brain connectivity and socio-emotional attributes likely exhibit complex, high-order interactions that linear methods like PCA cannot adequately capture.

As a result, the reduced feature representation lacked discriminative power for key classification subtasks. In particular, the sex classification task suffered severely, with the model completely failing to identify the female class (0.00 precision and recall), and the ADHD classification performance dropped in terms of balanced recall across classes. This suggests that PCA may have obscured important, task-specific features, especially for minority or more nuanced patterns in the data.

These findings highlight a critical limitation of PCA: while it is effective at preserving overall variance, it does not guarantee that the resulting components are optimal for predictive tasks. Consequently, this suggests a need to explore non-linear dimensionality reduction methods such as autoencoders, which proved to be infeasible to experiment with in this endeavor due to insufficient time and resources.

6.2 Discussion of Final Results

To evaluate the effectiveness of different machine learning approaches in classifying ADHD diagnosis, three models were compared: Random Forest, XGBoost, and an Artificial Neural Network. Performance was assessed using precision, recall, and F1 score, d the key metrics that capture different aspects of classification quality.

The ANN achieved the highest precision (0.72) among the models, indicating that when the model predicts a positive ADHD case, it is more likely to be correct. This is valuable in clinical contexts where false positives may lead to unnecessary follow-up or intervention. However, the ANN's recall was significantly lower (0.46), revealing that the model failed to identify a substantial proportion of true ADHD cases. As a result, its F1 score was 0.56, reflecting a moderate balance between precision and recall. In contrast, XGBoost demonstrated the strongest overall performance. It achieved perfect recall (1.00), meaning it successfully identified all ADHD cases in the dataset, while maintaining a respectable precision of 0.66. This resulted in the highest F1 score (0.80), making XGBoost the most balanced and effective model in this comparison.

Furthermore, the Random Forest model performed similarly to XGBoost, with slightly lower metrics: precision of 0.65, recall of 0.94, and F1 score of 0.77. While both tree-based models showed strong recall, XGBoost’s gradient boosting mechanism likely allowed for finer optimization of classification boundaries, giving it a marginal advantage. These results suggest that although the ANN demonstrated a stronger ability to avoid false positives, its lower recall undermined its overall classification effectiveness. This could stem from the ANN’s difficulty in generalizing over the high-dimensional input space or from insufficient training data. Conversely, ensemble tree-based methods like XGBoost and Random Forest proved more robust, especially in detecting positive ADHD cases, which is critical for minimizing underdiagnosis in sensitive applications like mental health screening.

6.3 Further Discussions

While this study focused on evaluating Random Forest, XGBoost, and an Artificial Neural Network (ANN) for ADHD diagnosis prediction, several avenues remain open for future research and model improvement. One key limitation relates to dimensionality reduction. Although Principal Component Analysis (PCA) was applied to reduce the high-dimensional input space (nearly 20,000 features), it did not significantly improve the model's performance—particularly for the ANN. PCA is a linear method and may not adequately capture the non-linear, complex relationships inherent in functional brain connectivity and behavioral features. To address this, future work could explore non-linear dimensionality reduction techniques such as autoencoders, which use a neural network architecture to learn compressed, task-relevant latent representations of the input data. By preserving more of the structural and relational complexity within the data, autoencoders could improve the model's ability to capture the most distinctive features.

Additionally, while the study focused on XGBoost and Random Forest as tree-based models, other ensemble methods could be investigated for comparative analysis. These include LightGBM, which offers faster training with large datasets, and CatBoost, which handles categorical variables more efficiently and often requires less data preprocessing. These models may offer performance gains or interpretability advantages, particularly in high-dimensional biomedical datasets like that of the functional connectome matrices dataset.

Finally, further hyperparameter optimization and smarter feature selection strategies may also enhance model generalizability. Given the challenges in predicting sex and ADHD concurrently, strategies such as task balancing, hierarchical modeling, or attention mechanisms could help improve the sensitivity of the ANN to underrepresented classes, like females in this dataset.

In summary, future work should aim to integrate more sophisticated dimensionality reduction methods and broaden the scope of classifiers to fully leverage the rich, complex features of functional neuroimaging and behavioral data in predictive modeling.

6.4 Alignment with the Requirements

The intelligent solution implemented—a multi-outcome neural network model for ADHD diagnosis and sex prediction—was developed to directly meet the client's requirements (WiDS Datathon 2025) as outlined on their home page (https://www.kaggle.com/competitions/widsdatathon2025)

The client's key needs included:

1. Accurate prediction of ADHD diagnosis using functional brain imaging data.
2. Inclusion of socio-demographic, emotional, and parenting factors to enhance prediction accuracy.
3. Simultaneous prediction of multiple outcomes (i.e., ADHD diagnosis and sex) from a unified model.
4. Interpretability and explainability to support research and clinical insights.

To address these, our team has designed a neural network that integrates both functional connectivity features from fMRI and tabular non-imaging data, using a dual-input architecture. The model processes each data modality through separate branches before merging them for final classification, ensuring that each data type is optimally represented. Here is how the solution aligns with each requirement:

Requirement 1: The model uses functional connectivity matrices derived from preprocessed fMRI data, allowing it to capture neural signatures associated with ADHD. Dimensionality reduction and regularization strategies were used to mitigate overfitting, which is common with high-dimensional neuroimaging data.

Requirement 2: The non-imaging inputs are encoded and processed through a fully connected sub-network, allowing these variables to inform the model alongside the neuroimaging features.

Requirement 3: A multi-output architecture was implemented, enabling the model to jointly predict ADHD diagnosis and sex. This not only meets the need for simultaneous outcome prediction but also takes advantage of potential shared patterns between the tasks to improve overall performance.

In summary, the proposed model aligns closely with the client's goals by offering a data-integrated, interpretable, and accurate solution for multi-outcome prediction in a neurodevelopmental context. It provides not only predictive power but also insights into the neural and psychosocial factors that contribute to ADHD, fulfilling both scientific and practical requirements.

7. Conclusion and Recommendation

In conclusion, future studies can build on the performance outcome measures of this paper, paving the way for future endeavors that seek to better understand women's health. This initiative is a small step to understanding the context of the problem, and there is still a long way to go. We suggest including more multi-faceted datasets that comprise more than the survey questions used for our dataset. This costs more energy and time, however it will prove to be fruitful in the long run, because there will be more different types of features with which to work.

Acknowledgements

We first and foremost are eternally indebted to Allah for allowing us to undertake this great initiative for the development of women's health in the field of science. We then gratefully acknowledge Dr. Rabab Alkhalifa for her invaluable guidance, insightful feedback, and unwavering support throughout every stage of this project. Her expertise greatly enhanced the rigor and clarity of this work. Finally, we would like to thank our parents for supporting us in this endeavor, and expecting the best of us in everything that we do.

References

[1] P. De Rossi, I. et al., "Gender-Related Clinical Characteristics in Children and Adolescents with ADHD," Journal of Clinical Medicine, vol. 11, no. 2, p. 385, Jan. 2022, doi: 10.3390/jcm11020385. Available: https://doi.org/10.3390/jcm11020385

[2] T. Suzuki et al., "Associations between adult Attention-Deficit/Hyperactivity Disorder (ADHD) traits and sociodemographic characteristics in Japanese workers," Neuropsychiatric Disease and Treatment, vol. Volume 19, pp. 759–773, Apr. 2023, doi: 10.2147/ndt.s398478. Available: https://www.dovepress.com/associations-between-adult-attention-deficithyperactivity-disorder-adh-peer-reviewed-fulltext-article-NDT

[3] F. D. Rabitho et al., "The Population of Children with High Risk of ADHD and The Mother's Sociodemographic Profile in Surabaya," International journal of scientific advances, vol. 5, no. 1, Jan. 2024, doi: https://doi.org/10.51542/ijscia.v5i1.8.

[4] G. M. S. El-Deen, A. M. Yousef, A. E. Mohamed, and A. S. Ibrahim, "Socio-demographic and clinical correlates of parenting style among parents having ADHD children: a cross-section study," Middle East Current Psychiatry, vol. 28, no. 1, Dec. 2021, doi: https://doi.org/10.1186/s43045-021-00151-3.

[5] S. Assari, "Emotional, Behavioral, and Cognitive Correlates of Attention Deficit and Hyperactive Disorder (ADHD) Screening and Diagnosis History: Sex/Gender Differences," Journal of Neurology & Neuromedicine, vol. 6, no. 1, pp. 1–12, Feb. 2021, doi: https://doi.org/10.29245/2572.942x/2021/1.1278.

[6] N. de Lacy and M. J. Ramshaw, "Identifying selective predictors of ADHD, Oppositional Defiant and Conduct Disorder onset in early adolescence with optimized deep learning," medRxiv (Cold Spring Harbor Laboratory), Aug. 2023, doi: https://doi.org/10.1101/2023.08.19.23294322.

[7] G. Argibay et al., "Predicting childhood and adolescent attention-deficit/hyperactivity disorder onset: a nationwide deep learning approach," Molecular Psychiatry, Dec. 2022, doi: https://doi.org/10.1038/s41380-022-01918-8.

[8] F. X. Castellanos, "A Biased Perspective on Brain Imaging of ADHD," American Journal of Psychiatry, vol. 178, no. 8, pp. 694–700, Aug. 2021, doi: https://doi.org/10.1176/appi.ajp.2021.21060609.

[9] H. W. Loh, C. P. Ooi, P. D. Barua, E. E. Palmer, F. Molinari, and U. R. Acharya, "Automated detection of ADHD: Current trends and future perspective," Computers in Biology and Medicine, vol. 146, p. 105525, Jul. 2022, doi: https://doi.org/10.1016/j.compbiomed.2022.105525.

[10] T. Chen, Ilias Tachmazidis, Sotiris Batsakis, Marios Adamou, E. P. Papadakis, and G. Antoniou, "Diagnosing attention-deficit hyperactivity disorder (ADHD) using artificial intelligence: a clinical study in the UK," Frontiers in Psychiatry, vol. 14, Jun. 2023, doi: https://doi.org/10.3389/fpsyt.2023.1164433.

[11] D. E. Attoe and E. A. Climie, "Miss. Diagnosis: A Systematic Review of ADHD in Adult Women," Journal of Attention Disorders, vol. 27, no. 7, pp. 645–657, Mar. 2023, doi: https://doi.org/10.1177/10870547231161533.

[12] S. Young et al., "Females with ADHD: An expert consensus statement taking a lifespan approach providing guidance for the identification and treatment of attention-deficit/ hyperactivity disorder in girls and women," BMC Psychiatry, vol. 20, no. 1, p. 404, Aug. 2020, doi: https://doi.org/10.1186/s12888-020-02707-9.

[13] R. Waite and M. Tran, "ADHD Among a Cohort of Ethnic Minority Women," Women & Health, vol. 50, no. 1, pp. 71–87, Mar. 2010, doi: https://doi.org/10.1080/03630241003601095.

[14] Scikit-Learn, "sklearn.ensemble.RandomForestClassifier — scikit-learn 0.20.3 documentation," *Scikit-learn.org*, 2018. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

[15] "XGBoost Documentation — xgboost 3.0.0 documentation," *Readthedocs.io*, 2025. https://xgboost.readthedocs.io/en/release_3.0.0/

[16] "Build the Neural Network — PyTorch Tutorials documentation," *pytorch.org*. https://pytorch.org/tutorials/beginner/basics/buildmodel_tutorial.html